

MINT-VLA: MILESTONE INTEGRATION WITH NATIVE-SPACE TARGETS FOR VISION-LANGUAGE-ACTION MODELS

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ABSTRACT

World-model-augmented robot policies condition actions on predicted future visual states, but latent targets can drift while a vision-language-action (VLA) encoder trains. We introduce **MINT-VLA** (Milestone Integration with Native-Space Targets for Vision-Language-Action models), which mines recurring stages, recomputes targets in the current encoder space, and injects one predicted prefix token. Held-out latent cosine improves from 0.7479 for persistence to 0.8134. Across three matched training seeds and 10,800 frozen-scene rollouts, no hint reaches 79.39%, offline absolute conditioning 75.44%, and MINT-VLA 70.81%. Their paired differences from no hint are -3.94 points (95% CI $[-6.58, -0.97]$) and -8.58 points ($[-11.47, -5.75]$). Correct predictions also fail to outperform current, zero, permuted, or cross-task controls, while a three-seed oracle-transition arm changes success by $+0.53$ points ($[-2.06, +3.00]$). Separately, a policy-preserving predictive adapter improves one fixed matched $\pi_{0.5}$ A0 by 11.61 points across three candidate seeds, adding 0.50% parameters and 0.93 ms direct-model latency. Independently matched A0 training reverses this conclusion: the three seed effects are $+13.42$, -5.50 , and -2.08 points, with mean $+1.94$ and 95% CI $[-5.78, +12.75]$. Its matched normal-shuffled effect is also only $+0.53$ points ($[-2.14, +3.08]$). Future-state predictability, representation alignment, and a one-seed gain therefore do not ensure replicated control utility.

1 INTRODUCTION

World models represent environment dynamics by predicting future observations or latent states from the current state and, in many formulations, an action. Recent vision-language-action (VLA) systems use predicted images, latent observation features, or visual subgoals to inform action generation (Hu et al., 2024; Black et al., 2024; Syed et al., 2026; Xu et al., 2026). Prediction quality alone, however, does not establish an improvement in control. WorldArena reports a correlation of only $r=0.360$ between world-model quality and planning performance (Shang et al., 2026), while VLA prediction systems use specialized attention, alignment, or gating before future supervision reaches the controller (Zhang et al., 2025; Xu et al., 2026; Zhu et al., 2026). The unresolved problem is therefore not only how to predict a future state, but how to integrate the prediction into a pretrained action policy.

This integration problem is especially important when the VLA visual encoder remains trainable. An independent visual world model preserves its own representation but repeats perception or generation at deployment. A precomputed latent is cheaper, yet an external encoder can organize the scene differently from the policy. Even a feature computed by the policy encoder becomes outdated as fine-tuning changes that encoder. Finally, an auxiliary future loss can alter shared current features whether or not the policy uses the predicted content. Encoder mismatch, target drift, forward conditioning, and backward representation shaping are different causes; a method-versus-baseline comparison cannot identify them.

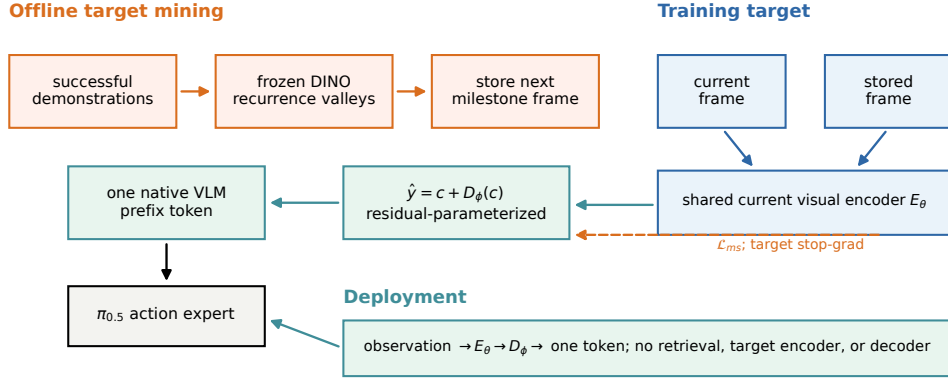


Figure 1: $\pi_{0.5}$ instantiation of MINT-VLA. Recurrence mining selects a later milestone frame from successful demonstrations. During training, the current policy encoder defines a stop-gradient target and a residual-parameterized predictor produces an absolute milestone feature. The $\pi_{0.5}$ adapter maps that feature to one prefix token.

Here we present **MINT-VLA** (Milestone Integration with Native-Space Targets for Vision–Language–Action models), a compact adapter that defines milestone targets in the policy’s current visual space (Fig. 1). MINT-VLA first finds stage states that recur across successful demonstrations and stores the corresponding frames. During policy training, the current encoder represents both the observation and a stop-gradient milestone target. A two-layer predictor estimates their displacement and reconstructs an absolute milestone feature. An architecture-specific projection passes this feature through the VLA’s native conditioning input; our $\pi_{0.5}$ implementation appends one prefix token. At deployment, the predictor operates on the current observation; it requires no retrieval, target encoder, or image decoder.

The completed experiments separate future-state predictability from control utility. On 480 held-out episodes, the predictor reaches latent cosine 0.8134 versus 0.7479 for persistence. Yet at fixed $\pi_{0.5}$ A2 and A3 checkpoints, correct future content fails to outperform current, zero, feature-permuted, cross-task, or within-task controls on 518 common scenes after Holm correction. Nine supporting six-task LaWAM comparisons give the same null result. Preserving a privileged milestone as a 4×4 patch grid does not repair the interface: it fails to beat both no-goal and current-image controls on Stack-3 and significantly increases the Hammer action endpoint error. Thus neither an empty predictor nor one-token spatial compression alone explains the control gap. An exact-paired public $\pi_{0.5}$ reaches 78.17% through the same evaluator. Across three matched corrected training seeds, no hint reaches 79.39%, offline absolute conditioning 75.44%, and MINT-VLA 70.81%. Relative to no hint, their paired hierarchical differences are -3.94 points (95% CI $[-6.58, -0.97]$) and -8.58 points ($[-11.47, -5.75]$), respectively. MINT-VLA is lower on all six tasks, while offline absolute conditioning improves only Stack-2 by 0.50 points. The corrected result therefore supports an integration diagnosis, not a method-benefit claim. A predeclared three-seed follow-up reaches the same boundary with privileged stage transitions: its mean effect over A0 is $+0.53$ points (95% CI $[-2.06, +3.00]$), with a negative effect at seed 1002 and regressions on Hammer and both ranking tasks.

We also evaluate a separate repair hypothesis that preserves the pretrained $\pi_{0.5}$ pathways and confines an action-conditioned predictive adapter to the action expert. Its preregistered seed-1000 screen improves over a matched current-source A0 by 13.42 points (82.42% versus 69.00%) without a task-level regression. The frozen P2 replication yields candidate effects of $+13.42$, $+9.08$, and $+12.33$ points across training seeds, with a mean $+11.61$ -point effect and hierarchical 95% interval $[+8.31, +14.67]$. This establishes candidate-seed replication relative to the fixed matched A0. Once A0 is independently trained at all three seeds, however, the matched effects are $+13.42$, -5.50 , and -2.08 points; their mean is $+1.94$ points

with hierarchical 95% CI $[-5.78, +12.75]$. The adapter also lacks content-specific causality: across three seeds the margin over shuffled action conditioning is $+0.53$ points (95% CI $[-2.14, +3.08]$; Holm-adjusted $p=0.534$). Adding recurrence-derived CRAVE targets reduces success to 62.92%; this rejects that auxiliary extension rather than the parent predictive adapter.

A separate demonstration-weighting screen gives a second bounded positive direction. At seed 1000, terminal-outcome weighting reaches 77.58%, versus 74.25% for ordinary fine-tuning and 71.08% for outcome-free CRAVE weighting. The predeclared three-seed replication retains positive mean differences of 2.81 and 3.81 points, but both hierarchical 95% intervals cross zero. We therefore reject a replicated utility claim rather than reinterpret these sample weights as a value or control-utility mechanism.

This paper makes three contributions:

- We introduce a milestone target recomputed by the current policy encoder and an adapter that uses each VLA’s native conditioning input; the evaluated $\pi_{0.5}$ implementation adds one token and 0.25% parameters.
- We define paired content interventions and a privileged spatial gate that distinguish prediction quality, content use, and interface capacity.
- We show across three matched training seeds that a nontrivial future predictor need not improve control: both conditioned arms regress, correct content has no resolved advantage, and a privileged oracle-transition route also fails its replicated closed-loop utility gate. A separately scoped policy-preserving adapter passes a candidate-seed gate against one fixed A0, but fails the independent matched-A0 gate and its content-specific intervention remains unresolved. Outcome-calibrated weighting also passes a one-seed directional screen, but its completed three-seed gate rejects a replicated method effect.

2 RELATED WORK

Predictive representations for robot policies. Prior systems predict future images, videos, latent observation features, or visual subgoals. VPP conditions inverse dynamics on future visual representations produced by a video diffusion model (Hu et al., 2024); SuSIE generates image subgoals for a goal-conditioned controller and compares them with ground-truth future images (Black et al., 2024). WorldVLA predicts future images together with actions, whereas DreamVLA forecasts dynamic, spatial, and semantic scene information for action prediction (Zhang et al., 2025; Cen et al., 2025). AHEAD predicts future patches in a frozen OpenVLA feature space (Syed et al., 2026). FutureVLA learns joint visuomotor representations and transfers them to downstream VLAs through latent alignment (Xu et al., 2026). Concurrent Δ VLA models changes relative to the current scene (Zhu et al., 2026). LaWAM predicts action-conditioned future observation features as latent visual subgoals; WLA-0 jointly predicts subtasks, images, and actions; and Enfold distills a generative world model into a representation computed from the current observation (Chen et al., 2026; Yang et al., 2026; Zeng et al., 2026). MINT-VLA therefore does not claim the first latent future state, visual subgoal, residual predictor, or auxiliary prediction loss. It addresses a specific integration problem: keeping target features consistent with a trainable policy encoder and mapping the prediction to the VLA’s existing conditioning interface.

Milestone selection. States shared across successful trajectories have long been used as bottlenecks or options (Şimşek & Barto, 2008; Sutton et al., 1999). MINT-VLA uses cross-episode visual recurrence to select candidate stage states without manually annotated boundaries. This differs from a fixed temporal horizon and from terminal-goal conditioning, but recurrence itself is not sufficient evidence that the selected state is useful. Our fixed-horizon, shuffled-action, masked-action, terminal, and oracle controls follow SuSIE’s principle of separating subgoal quality from controller capacity.

Integration and negative transfer. Auxiliary objectives can improve or damage a shared representation depending on weighting and task (Du et al., 2018; Lin et al., 2019; Yu

et al., 2020); negative transfer is not explained by gradient conflict alone (Jiang et al., 2023; Lyle et al., 2021). VLA systems accordingly use different interfaces, including structured attention, latent alignment, conditional variation modules, or a frozen policy feature space (Zhang et al., 2025; Xu et al., 2026; Zhu et al., 2026; Syed et al., 2026). Selective gradient routing is also used to preserve pretrained VLA semantics (Driess et al., 2025). MINT-VLA’s current implementation stops the milestone target branch but still allows its loss to update current visual features through the predictor. Because the matched utility gate fails, we do not attribute the regression to one gradient path; the content and privileged controls instead test whether the supplied condition is action-sufficient.

Establishing control benefit. Prediction quality and control performance can diverge (Shang et al., 2026; Jiang et al., 2026; Yu et al., 2026), and correlated conditioning can induce causal confusion (de Haan et al., 2019; Wen et al., 2020). Evaluation seeds also do not replace independent training seeds (Henderson et al., 2018; Agarwal et al., 2021). We therefore require closed-loop success, fixed-checkpoint content interventions, task-level reporting, and training-seed replication. These controls are necessary to determine whether performance depends on the predicted future state rather than merely on an additional visual token.

3 PROBLEM SETTING

Let $h_t = E_\theta(o_t)$ be the policy representation, y_t a future milestone feature, and q_t the condition presented to the action policy:

$$a_t = \pi_\psi(h_t, q_t, l), \quad (1)$$

where l is the instruction. A future-conditioned VLA couples four choices: which future frame defines the target; whether its feature comes from an external, frozen, or current encoder; whether the policy receives an absolute feature or displacement; and whether the auxiliary gradient updates shared features. A projection aligns dimensions but not semantic neighborhoods, and cached targets cannot follow a trainable encoder.

These choices require distinct evidence. Matched no-condition training tests overall utility; fixed-checkpoint zero and shuffled controls test route use and content use; privileged semantic and spatial targets separate predictor error from interface capacity; independent training seeds measure optimization robustness; and a mature public checkpoint bounds recovery from weak fine-tuning. Simulator seeds estimate rollout variation for one checkpoint and cannot replace training seeds. LaWAM and LIBERO are supporting diagnostics because the former lacks a clean VLA-from-base comparison and the latter is near saturation for the studied policies.

4 MINT-VLA

4.1 OVERVIEW

MINT-VLA separates future-state supervision from the mechanism used by a particular VLA to receive additional conditioning. It has four components: (i) milestone images selected from successful demonstrations, (ii) target features computed by the VLA’s current visual encoder, (iii) a predictor that estimates a milestone feature from the current observation, and (iv) a small architecture-specific adapter that passes this feature to the policy. The demonstrations and milestone images are used only for training. Deployment retains the policy encoder, milestone predictor, and adapter.

4.2 MILESTONE TARGET DISCOVERY

Successful demonstrations repeatedly pass through recognizable task stages: an object is grasped, one block is placed, or a handover is completed. For a frame o_t^i from episode i , we

L2-normalize a frozen DINOv3 pooled feature $f(o_t^i)$ and compute

$$r(o_t^i) = \frac{1}{|\mathcal{E}| - 1} \sum_{j \neq i} \max_s \exp\left(-\frac{\|f(o_t^i) - f(o_s^j)\|_2^2}{2\sigma^2}\right), \quad (2)$$

where σ is the median cross-episode nearest-neighbor distance. We Gaussian-smooth each episode’s score ($\sigma_t=1.4$), split it at valleys with prominence 0.03 and temporal separation $\max(2, \lfloor T/12 \rfloor)$, and select the highest-scoring frame in each segment. Each frame is paired with the representative of the next segment, or with the final frame after the last boundary. We process at most 200 uniformly sampled successful episodes per task. This procedure uses task grouping and success labels, but no manually annotated stage boundaries.

4.3 NATIVE-SPACE TARGET CONSTRUCTION

Let E_θ denote the visual encoder of the VLA being trained. MINT-VLA stores the milestone image o_t^{ms} , rather than a feature computed before policy training. At each update, the current encoder maps the observation and milestone image into the same representation space:

$$c_t = \text{pool}(E_\theta(o_t)), \quad y_t = \text{sg}[\text{pool}(E_\theta(o_t^{\text{ms}}))]. \quad (3)$$

We call y_t a *native-space target* because its coordinates are defined by the current policy encoder, not by an external encoder or an earlier encoder snapshot. Stop-gradient prevents the target branch from changing E_θ , while allowing the target itself to follow changes in the encoder over training.

4.4 MILESTONE PREDICTION

A two-layer predictor estimates a change from the current feature and reconstructs an absolute milestone representation:

$$\hat{y}_t = c_t + D_\phi(c_t), \quad \mathcal{L}_{\text{ms}} = \|\hat{y}_t - y_t\|_2^2. \quad (4)$$

Thus, “residual” describes the predictor parameterization; the policy receives the absolute prediction $\hat{y}_t$. The current predictor is conditioned on the observation, while the task instruction is supplied separately to the VLA. This design cannot distinguish instruction-dependent futures that share the same visual observation. Moreover, stop-gradient protects only the target branch: $\nabla_\theta \mathcal{L}_{\text{ms}}$ can still pass through c_t . We therefore do not describe the current method as gradient-isolated.

4.5 POLICY-SPECIFIC CONDITIONING ADAPTER

The predicted milestone is connected to a policy through an architecture-specific adapter A_ω :

$$q_t = A_\omega(\hat{y}_t), \quad a_t = \pi_\psi(o_t, l, q_t), \quad (5)$$

where l is the instruction and q_t follows the conditioning format exposed by the VLA. For a token-based policy, A_ω projects the milestone into a conditioning token. A policy with an existing cross-attention or latent conditioning input can instead map $\hat{y}_t$ to that interface. Milestone selection, native-space supervision, and the residual prediction objective are shared across implementations. The visual encoder and adapter come from the target VLA, and the predictor is sized to that encoder’s feature width. This separation makes MINT-VLA compatible by design with multiple VLA interfaces, but the present experiments establish performance only for the $\pi_{0.5}$ instantiation. We do not claim empirical transfer to a second architecture; that expansion was closed after the primary positive-utility gate failed.

4.6 $\pi_{0.5}$ INSTANTIATION

$\pi_{0.5}$ contains a PaliGemma visual-language prefix and an action expert. We implement A_ω as a linear projection to the prefix width and append one bidirectionally visible milestone token:

$$q_t = P_\omega(\hat{y}_t), \quad X_{\text{prefix}} = [X_{\text{image}}, q_t, X_{\text{text}}]. \quad (6)$$

The milestone-loss weight is 0.05. This implementation adds no cross-attention tower, second visual encoder, image decoder, or demonstration retrieval at deployment. It adds 0.25% parameters to the evaluated $\pi_{0.5}$ policy. The A0–A3 comparison arms and their matched training configuration are defined in Section 5; complete implementation values are listed in Appendix A.

5 EXPERIMENTAL DESIGN

Benchmark and pairing. The main experiment uses six RoboTwin 2.0 tasks spanning contact, ranking, handover, and two- or three-stage stacking. Each accepted checkpoint is evaluated on four simulator seeds and 50 frozen scenes per task–seed cell (1,200 episodes). Contrasts pair identical scene and episode keys. Training seed is the top-level uncertainty unit in a hierarchical bootstrap; paired episodes are resampled within each equally weighted task. We report every task and require the 95% interval for the macro difference to exclude zero. A task-safe claim additionally permits no seed–task regression below -5 points.

Matched policy comparison. The corrected $\pi_{0.5}$ recipe uses absolute actions, mean/std normalization, batch 16, 50,000 updates, and the published optimizer schedule. At training seeds 1000–1002 we compare A0 (no milestone), A2-Abs (an offline absolute milestone token), and A3 (the current-encoder MINT-VLA token). All nine final step-49,999 checkpoints pass source, initialization, dataset, normalization, optimizer, and evaluator-manifest audits. This matrix is the primary utility test; earlier joint-delta, batch-64, 20k-update runs are exploratory only.

Fixed-checkpoint content controls. To ask whether action depends on the predicted future rather than an extra route or correlated token, we replace the correct condition with current, zero, feature-permuted, cross-task, and within-task content while holding the checkpoint and scenes fixed. A separate three-seed adapter panel compares normal inference with shuffled actions, a zeroed route, and masked actions, with Holm correction over the three preregistered mechanism contrasts.

Privileged upper bounds. Three controls remove distinct possible bottlenecks before training further predictors: same-scene oracle stage transitions, explicit correct-next-subtask prompts on a frozen public policy, and a ground-truth 4×4 spatial milestone in a two-task offline action probe. Each has a frozen no-condition or current-condition baseline and a content-shuffled control where applicable. Failure stops the corresponding tracker, semantic predictor, or spatial predictor expansion.

Policy-preserving predictive branch. This independently designed branch leaves inherited $\pi_{0.5}$ visual-language and action paths intact, stops predictive gradients at inherited features, and routes a zero-initialized action-conditioned adapter only into the action expert. P1 screens seed 1000 against a current-source A0 and route/action controls. P2 trains candidate seeds 1001 and 1002 against the fixed seed-1000 A0. P3 supplies the decisive denominator by independently training matched A0 seeds 1001 and 1002; only the three seed-matched comparison can support a replicated method claim. P4 tests content, route, and action-conditioning use, and P5 compares the candidates with the exact public checkpoint. The adapter cost is measured by parameter count, compiled cost, memory, direct inference, WebSocket latency, and throughput.

Predeclared stopping. Recurrence-aligned auxiliary training and outcome-calibrated demonstration weighting are evaluated as separate extensions. A one-seed directional screen may authorize replication but cannot establish a method effect. New task panels, a second VLA, or component-factorization training are launched only if their upstream utility gates pass. Complete configurations, intervention identities, multiplicity families, and audit hashes are in Appendix A.

Table 1: Corrected three-seed RoboTwin result. Each method contains 3,600 paired episodes. Task entries are mean percentage-point effects versus the matched A0; intervals resample training seeds and paired scenes within task.

Method	Success	Hammer	Rank RGB	Rank size	Handover	Stack-2	Stack-3	Δ A0 (95% CI)
A0	79.39	—	—	—	—	—	—	—
A2-Abs	75.44	-3.17	-5.83	-6.00	-5.50	+0.50	-3.67	-3.94 [-6.58, -0.97]
MINT-VLA	70.81	-8.67	-6.50	-13.00	-8.83	-2.83	-11.67	-8.58 [-11.47, -5.75]

6 RESULTS

6.1 CORRECTED TRAINING REJECTS THE MILESTONE INTERFACES

The published $\pi_{0.5}$ checkpoint reaches 78.17% on our frozen six-task, four-seed RoboTwin evaluator (1,200 episodes), showing that the bridge supports a mature policy. It is a calibration rather than a matched baseline because its training mixture differs. We therefore train A0 (no hint), A2-Abs (offline absolute milestone), and A3 (MINT-VLA) from the same official initialization under the corrected public recipe at seeds 1000–1002.

Across 10,800 rollouts, A0 reaches 79.39%, A2-Abs 75.44%, and A3 70.81%. The matched hierarchical differences are -3.94 points for A2-Abs (95% CI [-6.58, -0.97]) and -8.58 for A3 ([-11.47, -5.75]). A3 regresses on all six tasks; A2-Abs improves only Stack-2 by 0.50 points. Thus the internally matched but weak legacy pilot, where every conditioned arm improved over A0, does not survive the corrected training protocol. Full seed-by-task success rates and the legacy matrix are reported in Appendix A.

6.2 PREDICTION QUALITY AND PRIVILEGED TARGETS DO NOT IMPLY ACTION UTILITY

The predictor is nontrivial: over two task-stratified held-out folds (480 episodes), latent cosine is 0.8134 versus 0.7479 for persistence, with 46.9% top-1 and 80.8% top-5 milestone retrieval. At fixed A2 and A3 checkpoints, however, correct predictions do not outperform current, zero, feature-permuted, cross-task, or within-task replacements on the common 518 scenes; all pooled Holm-adjusted p -values equal 1. Nine six-task LaWAM comparisons give the same null result, with one corrected task-level difference and it is adverse.

Three privileged controls further isolate the failure. First, oracle stage transitions change success by only +0.53 points over matched A0 across three training seeds (95% CI [-2.06, +3.00]). Second, a correct next-subtask prompt reaches 77.08% on a fixed public policy, below no subtask at 80.83% and statistically indistinguishable from shuffled semantics; Handover regresses by 25 points. Third, a ground-truth 4×4 milestone grid does not beat both no-goal and current-image controls in the two-task offline action probe and significantly worsens Hammer endpoint error. These tests remove predictor error, semantic ambiguity, or global pooling in turn, but none establishes that the future condition is action-sufficient.

6.3 A POLICY-PRESERVING ADAPTER ALSO FAILS INDEPENDENT REPLICATION

A separately scoped predictive adapter preserves inherited visual-language and action paths and routes a zero-initialized, action-conditioned branch only to the action expert. Its seed-1000 screen improves over matched A0 by 13.42 points without a task regression. Replicating only candidate training against that fixed A0 also appears positive (+11.61 points; 95% CI [+8.31, +14.67]), but this denominator omits baseline-training variation.

P3 independently trains the missing A0 seeds. The matched effects become +13.42, -5.50, and -2.08 points (Table 2); their mean is +1.94 with hierarchical 95% CI [-5.78, +12.75]. The primary gate fails, as does task safety. An exact-paired mature public reference likewise gives only +2.44 points ([-0.72, +5.39]). Across candidate seeds, normal minus shuffled actions is +0.53 points ([-2.14, +3.08]); zero-route and masked-action contrasts also cross zero

Table 2: P3 independent matched-baseline replication. Each seed pairs 1,200 candidate and A0 episodes. Task columns are candidate-minus-A0 percentage points; Effect is their equally weighted macro.

Seed	A0	Candidate	Hammer	Rank RGB	Rank size	Handover	Stack-2	Stack-3	Effect
1000	69.00	82.42	+8.5	+13.5	+15.5	+14.0	+2.0	+27.0	+13.42
1001	83.58	78.08	-4.0	-1.0	-13.0	-1.5	-4.5	-9.0	-5.50
1002	83.42	81.33	+2.0	-5.5	-5.0	-3.0	+0.5	-1.5	-2.08
Mean	78.67	80.61	+2.17	+2.33	-0.83	+3.17	-0.67	+5.50	+1.94

after Holm correction. The fixed-A0 screen therefore establishes neither an independently replicated method effect nor use of correct predicted content.

The adapter adds 16.82 million parameters (0.50%). Direct-model latency rises by 0.93 ms (1.63%), WebSocket round-trip latency by 2.76 ms (2.48%), and matched four-GPU peak training memory by 18 MiB (0.027%). These modest costs do not rescue the rejected utility and mechanism gates.

6.4 ADDITIONAL EXTENSIONS REMAIN NEGATIVE

Adding recurrence-derived CRAVE targets to the adapter lowers seed-1000 success to 62.92%, 6.08 points below A0. Terminal-outcome demonstration weighting passes a one-seed directional screen, but across three seeds its mean gains are only +2.81 points over ordinary training (95% CI $[-0.94, +6.58]$) and +3.81 over outcome-free CRAVE weighting ($[-0.25, +7.97]$). Both replication gates therefore fail. Correlated recurrence statistics remain useful diagnostics, not evidence for a reward, value function, or model-predictive controller.

The three independent negative branches close the preregistered selector, tracker, task-expansion, second-VLA, and component-factorization paths. We do not tune new variants against these outcomes. The evidence is restricted to the evaluated $\pi_{0.5}$ interfaces and RoboTwin panel; complete controls, per-seed task tables, audits, and scope qualifications are in the appendix.

7 DISCUSSION AND CONCLUSION

MINT-VLA improves held-out future-feature prediction, but the corrected three-seed control experiment rejects action utility: A2-Abs and MINT-VLA are -3.94 and -8.58 points below matched no-hint training, respectively. Correct content also fails to beat fixed-checkpoint replacements. Oracle stage transitions, explicit next-subtask semantics, and privileged spatial targets do not reverse the result. A later visual state can identify progress without specifying path, contact, arm assignment, or timing, so predictability and representation alignment are not sufficient evidence of action sufficiency.

Preserving pretrained pathways does not yield a stable exception. The predictive adapter has a strong seed-1000 screen and a positive comparison with one fixed A0, but independent matched A0s change the three seed effects to +13.42, -5.50, and -2.08 points (mean +1.94; 95% CI $[-5.78, +12.75]$). Its correct-versus-shuffled, route, mature-public, and task-safety gates also fail. Recurrence-aligned auxiliary training and outcome-calibrated weighting provide further negative replication results.

The conclusion is deliberately scoped to the tested $\pi_{0.5}$ milestone and predictive-adapter interfaces and the six-task RoboTwin panel. It does not rule out generative, action-conditioned, spatial, or planning-oriented world models. It does show that latent prediction metrics, one-seed gains, and extra conditioning routes must not substitute for matched training seeds and causal content controls.

REPRODUCIBILITY STATEMENT

The anonymized supplementary package includes preprocessing and evaluation code, effective training configurations, checkpoint identifiers, and aggregate evaluation reports. Appendix A records source and data hashes, software versions, simulator seeds, and the currently available artifact coverage. The accepted-scene manifest, launch records, per-episode outcomes, and paired three-seed report accompany the confirmatory matrix so that all contrasts can be reproduced. The oracle-transition follow-up additionally includes three per-seed reports and its failed-gate decision JSON. The recurrence rollout diagnostic and semantic-subtask screen include artifact audits, frozen protocols, per-episode outcomes, and rejected-gate records. The policy-preserving predictive branch additionally includes the five P1 seed-1000 reports, the accepted-gate JSON, both P2 candidate reports, the fixed-A0 candidate-seed gate, both independently trained P3 A0 reports and checkpoint audits, the rejected P3 matched-seed gate, the matched efficiency report, the four R1 extension reports, and the rejected R1 gate. The outcome-calibrated branch includes all nine three-arm reports, the accepted seed-1000 screen, and the rejected three-seed replication gate. Replicated claims are limited to branches whose frozen gate completed; rejected branches remain reported as negative results.

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AI USAGE DISCLOSURE

Generative AI tools assisted literature discovery, hypothesis refinement, experimental design, implementation and code review, figure and table formatting, drafting and translation, and interpretation of experimental results. They were not used to generate demonstrations, simulator outcomes, or reported measurements. The authors inspected the relevant source code, checked configuration and artifact provenance, ran builds and tests, and verified paper claims against the underlying reports. The authors take responsibility for all content and artifacts in this submission.

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A APPENDIX

A.1 EFFECTIVE CONFIGURATIONS

The completed A0–A3 pilot shares `pi05_base`, joint-delta actions, quantile normalization, 20k updates, batch size 64, and a 50-action horizon. A0 sets `lmwm_hint_dim=0`. A1 and A2 use one prefix token with dimensions 768 and 1152. A3 uses dimension 1152, `lmwm_live_hint=True`, `lmwm_live_residual=True`, and milestone-loss weight 0.05. The confirmatory configuration instead uses absolute actions, mean/std normalization, batch 16, 50k updates, and the published optimizer schedule. The audited configuration source has SHA-256 `fe9377c2b93f2428...4dd7264`; the corrected A0 config is `pi05_robotwin_a0_public_recipe_bj`. It disables joint deltas and quantile normalization, uses the `robotwin2.0_absolute_meanstd` statistics, AdamW weight decay 0.01, batch 16, and 50k updates. Every confirmatory arm records its resolved config and dataset-manifest hash next to its checkpoint rather than relying on the mutable source file. All nine A0/A2-Abs/A3 checkpoints at training seeds 1000–1002 finalize step 49,999 and pass immutable launch, source, checkpoint, dataset, and evaluator audits. The accepted scene-manifest SHA-256 is `08ed8eb7fa7e166e...9418b820d17`; all protocol checks pass with no failures.

A.2 EVALUATION AND STATISTICS

Pilot simulator seed groups begin at 100k, 200k, 300k, and 400k. Candidate initializations are retained only when the expert planner and benchmark success check both pass. Each accepted cohort has 50 episodes per task and seed group. Because filtering can produce different accepted scenes, intervention tests use the intersection of scene identifiers. The completed six-task diagnosis contains 1,200 paired scenes per method–control comparison and records discordant counts, exact McNemar tests, and Holm-adjusted p -values in `logs/l2_six_task_intervention_analysis.json`. Confirmatory training reports include seed-level task cells and hierarchical bootstrap intervals. Independently filtered cohorts are never treated as matched observations.

Each of the nine accepted evaluations uses the frozen confirmatory manifest and contains all 24 task–evaluation–seed cells. Seed-level macros are A0: 75.50/80.00/82.67%, A2-Abs: 70.00/74.67/81.67%, and A3: 66.92/68.92/76.58%. Across training seeds, A0 task means are 89.50 Hammer, 95.00 ranking-RGB, 61.83 ranking-size, 59.83 Handover, 92.83 Stack-2, and 77.33 Stack-3. A2-Abs changes these by -3.17 , -5.83 , -6.00 , -5.50 , $+0.50$, and -3.67 points; A3 changes them by -8.67 , -6.50 , -13.00 , -8.83 , -2.83 , and -11.67 points. The paired hierarchical macro differences are -3.94 points for A2-Abs (95% CI $[-6.58, -0.97]$) and -8.58 points for A3 ($[-11.47, -5.75]$), each over 3,600 paired episodes with zero unmatched episode keys.

The oracle-transition follow-up uses the same three A0 reports and adds three audited MT1 reports, each with 24 cells and 1,200 episodes. Oracle macros are 77.75/81.00/81.00%; paired differences from A0 are $+2.25$ / $+1.00$ / -1.67 points. Across training seeds, task differences are -1.50 Hammer, -0.83 ranking-RGB, -1.50 ranking-size, $+1.67$ Handover, $+2.17$ Stack-2, and $+3.17$ Stack-3. The paired hierarchical macro difference is $+0.53$ points (95% CI

$[-2.06, +3.00]$) over 3,600 paired episodes with zero unmatched keys. The interval crosses zero, so the predeclared MT1 gate rejects the oracle-transition branch.

A.3 POLICY-PRESERVING PREDICTIVE-ADAPTER SCREENS

P1 trains matched current-source A0 and predictive checkpoints from the same official $\pi_{0.5}$ initialization at seed 1000. Both finalize step 49,999 and pass source, dataset, normalization, parameter-tree, optimizer-state, checkpoint, and 24-cell scene-manifest audits. Every condition contains 1,200 paired episodes. Macro success is 69.00% for A0, 82.42% for normal predictive inference, 78.50% with the predictive route zeroed, 81.17% with shuffled action conditioning, and 78.00% with masked actions. Normal minus A0 is +13.42 points, with task effects of +8.5, +13.5, +15.5, +14.0, +2.0, and +27.0 points in table order. The pooled Holm-adjusted A0 contrast is 7.02×10^{-16} . Normal minus shuffled actions is only +1.25 points (exact McNemar $p=0.383$; Holm-adjusted $p=1.0$), so the accepted directional P1 gate does not establish content-specific causality or multi-seed utility. The canonical decision is `RESULTS_pi05_predictive_adapter_p1_seed1000_gate.json`.

P2 completes the frozen normal-candidate replication at training seeds 1001 and 1002. Together with seed 1000, candidate macros are 82.42%, 78.08%, and 81.33%, giving effects of +13.42, +9.08, and +12.33 points against the fixed seed-1000 matched A0. The equal-seed mean effect is +11.61 points; the predeclared hierarchical paired 95% interval is $[+8.31, +14.67]$. Mean task effects in Hammer, ranking-RGB, ranking-size, Handover, Stack-2, and Stack-3 order are +4.50, +10.67, +11.67, +18.50, +1.33, and +23.00 points. The fixed-A0 candidate-seed gate is accepted. P4 repeats the normal-shuffled, zero-route, and masked-action interventions across all three candidate seeds, but their mean effects are only +0.53, +1.50, and +1.33 points; all hierarchical 95% intervals cross zero and all Holm-adjusted paired tests exceed 0.20. The mechanism gates therefore fail. Because P2 holds the A0 training seed fixed, it establishes candidate-seed replication relative to that A0, not content-specific causality or independent baseline-seed replication.

P3 independently trains the missing A0 seeds. A0 macros at seeds 1000–1002 are 69.00%, 83.58%, and 83.42%, versus candidate macros of 82.42%, 78.08%, and 81.33%. The matched effects are therefore +13.42, -5.50, and -2.08 points; their equal-seed mean is +1.94 points and the hierarchical 95% interval is $[-5.78, +12.75]$. The primary gate rejects independent matched-seed utility. Task effects for seeds 1000/1001/1002, in Hammer, ranking-RGB, ranking-size, Handover, Stack-2, and Stack-3 order, are (+8.5, +13.5, +15.5, +14.0, +2.0, +27.0), (-4.0, -1.0, -13.0, -1.5, -4.5, -9.0), and (+2.0, -5.5, -5.0, -3.0, +0.5, -1.5) points. The task-safety gate also fails. The canonical records are `RESULTS_pi05_predictive_adapter_p2_gate.json` and `RESULTS_pi05_predictive_adapter_p2_efficiency.json`, together with `pi05_predictive_adapter_p3_matched_seed_gate.json`.

P5 reevaluates the exact public checkpoint on the same 24 cells and recovers all 1,200 episode identities. It obtains 78.17% success. Candidate-minus-public effects are +4.25, -0.08, and +3.17 points, with equal-seed mean +2.44 points and hierarchical 95% CI $[-0.72, +5.39]$. The interval crosses zero, so the mature-initialization gate is rejected. The canonical record is `RESULTS_pi05_predictive_adapter_p5_public_gate.json`.

The matched efficiency probe counts 3.353 and 3.370 billion parameters (+16.82 million; 0.50%). Direct-model latency is 57.10 versus 58.03 ms, WebSocket round-trip latency is 111.57 versus 114.33 ms, and throughput is 8.96 versus 8.75 requests/s over five warmups and 30 trials on the same GPU. Compiled XLA cost analysis reports 172.29 versus 164.92 GFLOPs, which is a compiler estimate rather than a theoretical operation count. The matched four-GPU ten-step training probe peaks at 66,789 versus 66,807 MiB (+18 MiB).

R1 uses the corrected 359,823-row dense recurrence-target protocol. Its CRAVE-only, predictive-plus-CRAVE, combined zero-route, and combined shuffled conditions reach 68.67%, 62.92%, 62.17%, and 64.67%, respectively, over 1,200 paired episodes each. Predictive-plus-CRAVE is 6.08 points below A0 (paired bootstrap 95% CI $[-9.33, -2.75]$), 5.75 points below CRAVE-only ($[-9.00, -2.50]$), and has four task regressions larger than

five points. The rejected R1 gate disables its additional seeds. Because R1 adds an auxiliary target to P1, this verdict does not negate the parent adapter or its completed P2 replication.

A.4 OUTCOME-CALIBRATED WEIGHTING SCREEN

R4 uses the same 6,313 expert-supervised query-action chunks in three matched seed-1000 arms: ordinary unit weighting, task-normalized outcome-free CRAVE progress-change weighting, and aligned-query-rollout terminal-outcome weighting. Each fixed step-5,000 checkpoint passes the frozen 24-cell, 1,200-episode audit. Macro success is 74.25%, 71.08%, and 77.58%, respectively. Terminal minus ordinary is +3.33 points (paired hierarchical 95% CI $[-2.25, +8.92]$); terminal minus outcome-free CRAVE is +6.50 points ($[+0.33, +12.92]$).

Terminal-minus-ordinary task effects in Hammer, ranking-RGB, ranking-size, Handover, Stack-2, and Stack-3 order are +6.0, +0.5, +8.0, +8.5, -3.5, and +0.5 points. Terminal-minus-CRAVE effects are +8.5, +3.0, +4.5, +15.5, -2.0, and +9.5. The directional gate accepts the arm because both macro point estimates improve and no task regresses by more than five points. The crossing-zero ordinary contrast prevents a significance or replicated-effect claim. The canonical records are `RESULTS_pi05_r4_seed1000_complete.json` and `RESULTS_pi05_r4_seed1000_gate.json`; the frozen seed-1001/1002 replication was therefore required. These weights do not constitute Q-values, action advantages, rewards, a world critic, or model-predictive control.

The frozen replication is now complete. At seeds 1000, 1001, and 1002, terminal-minus-ordinary macro effects are +3.33, +0.08, and +5.00 points; terminal-minus-CRAVE effects are +6.50, +0.58, and +4.33 points. The three-seed mean effects are +2.81 points against ordinary (hierarchical 95% CI $[-0.94, +6.58]$) and +3.81 against outcome-free CRAVE ($[-0.25, +7.97]$). Mean task effects against ordinary, in Hammer, ranking-RGB, ranking-size, Handover, Stack-2, and Stack-3 order, are +1.67, +1.67, +6.67, +5.33, +0.50, and +1.00 points. Against CRAVE they are +3.00, +2.50, +1.83, +7.17, +2.00, and +6.33 points. No seed-task contrast falls below -5 points, but neither interval excludes zero; the predeclared replication gate is therefore rejected. The canonical record is `RESULTS_pi05_r4_three_seed_gate.json`.

A.5 OUTCOME-LINKED RECURRENCE DIAGNOSTICS

The frozen recurrence-diagnostic panel contains 120 public- $\pi_{0.5}$ rollouts: 99 successes and 21 failures across the six RoboTwin tasks. All 12 task-simulator-seed summaries and 120 videos pass the artifact audit. Successful-minus-failed terminal progress is 0.11129 (episode-bootstrap 95% CI $[0.04397, 0.18444]$) and progress gain is 0.16166 ($[0.08935, 0.24513]$). Failed-minus-successful stall fraction is 0.34827 ($[0.30419, 0.39356]$) and regression fraction is 0.12523 ($[0.02950, 0.22978]$). These separations authorize a causal readout test, not a reward or value claim. Low density, stall, and regression have success false-positive rates of 98.99%, 89.90%, and 38.38%, respectively; ranking RGB has no failed rollout in this panel. The canonical record is `logs/crave_r0/rollout_analysis/report.json`.

A.6 PRIVILEGED SEMANTIC-SUBTASK PROTOCOL

The semantic screen keeps the public $\pi_{0.5}$ checkpoint, tokenizer, 24-cell scene manifest, replan horizon, and action budget fixed across five conditions. Four evaluation seeds contribute 10 episodes per task, for 240 paired episodes per condition. The primary gate requires the paired 95% lower bound for semantic-next minus no-subtask to exceed zero; semantic next must also exceed generic-stage, semantic-current, and within-task shuffled semantics in point estimate, with no task regression below -5 points. The observed semantic-next minus no-subtask difference is -3.75 points (95% CI $[-10.42, +2.92]$), and semantic-next minus shuffled is -0.42 points ($[-7.08, +6.25]$). Handover regresses by 25.0 points. The gate is rejected; the protocol and final analysis are recorded in `manifests/pi05_r3_semantic_screen_protocol_v1.json` and `logs/r3_semantic_screen/report.json`.

A.7 RECURRENCE MINING

The mining script (SHA-256 d05eefc2f9690163...62c) applies the following deterministic procedure independently per task: (1) uniformly retain at most 200 successful episodes; (2) L2-normalize pooled DINOv3 features; (3) set the RBF bandwidth to the median cross-episode nearest-neighbor distance; (4) compute the average nearest-match recurrence score; (5) smooth with a temporal Gaussian of width 1.4; (6) detect valleys with prominence 0.03 and minimum distance $\max(2, \lfloor T/12 \rfloor)$; (7) choose the maximum-recurrence frame inside each resulting segment; and (8) pair every frame with the next segment’s representative, using the terminal frame in the final segment. Valley distance acts as temporal NMS, and selecting one ridge per segment performs within-episode deduplication. Cross-episode similarity enters the score rather than a separate global prototype-clustering stage.

An artifact audit exposed incomplete pilot coverage. The current `pairs.npz` (SHA-256 5d68e21fbf0dd462...8ade8) contains 335,624 pairs from five tasks and omits Stack-3 entirely:

Task	Episodes	Pairs	Unique targets	Future pairs
Hammer	200	22,820	260	99.12%
Stack-2	200	82,204	652	99.76%
Ranking RGB	200	91,836	829	99.78%
Ranking size	200	91,932	800	99.78%
Handover	200	46,832	339	99.57%
Stack-3	0	0	0	—

When a pair is absent, the data transform supplies the current image as a shape-compatible placeholder and sets the milestone-loss mask to false. The A3 predictor token remains active through the action loss, but receives no milestone auxiliary supervision for that sample. Consequently the legacy Stack-3 A3 cell measures an added current-state predictor/capacity path, not a recurrence-supervised milestone. We therefore generated the frozen `robotwin_milestone_all16_confirmatory_v1` artifact. It contains 200 episodes for each of the six tasks (1,200 total), including 94,141 Stack-3 pairs; its `pairs.npz` SHA-256 begins 592f1e739105951e, and its READY audit SHA-256 begins 4353c725f7e55f9c. It passes nonzero coverage, monotone-target, selection-manifest, and source-hash checks and is the only recurrence artifact allowed for confirmatory A3 training. Threshold sensitivity and alternative selectors would require newly trained policies, but the positive-utility prerequisite failed and these expansions are closed.

A.8 LAWAM MECHANISM PROBE

All diagnostic variants inherit `lawam_pretrain`, which contains future prediction and distillation, and retain current DINO conditioning. The former “no-WM” row is therefore *LaWAM-init/Future-off*, not a clean VLA baseline. Across replicated training seeds, Future-off averages 90.83% (three seeds), whereas Local future, Absolute, Residual, Absolute+stop-gradient, and Residual+stop-gradient obtain 86.67, 89.21, 87.50, 89.04, and 88.78%, respectively. The latter values use two seeds except Residual+stop-gradient, which uses three. Future-off inherits LaWAM-aware pretraining and therefore does not identify a clean VLA pretraining effect.

At fixed checkpoints, all nine pooled correct-versus-zero, within-task, and cross-task comparisons have Holm-adjusted $p=1$. The adverse Handover contrast for Residual+stop-gradient is 72.5% correct versus 82.5% with a same-task foreign-instance hint (Holm-adjusted $p=0.0337$). These data diagnose downstream future conditioning but do not replace the corrected $\pi_{0.5}$ A0/A2/A3 comparison.

A.9 EXPLORATORY TASK GROUPS AND LIBERO BOUNDARY

In the legacy Stack-2/3 subset, A0, A2-Abs, A2-Res, and A3 average 29.75, 49.75, 40.75, and 54.75%. The public no-hint checkpoint reaches 81.75%. This post-hoc two-task pattern is

not a validated applicability law. The predeclared extension covers block and bowl stacking and includes reactive and geometric controls.

The auxiliary $\pi_{0.5}$ LIBERO diagnostic uses checkpoint step 29,999, OSMesa rendering, three rollout seeds, and the same one-prefix-token interface:

Suite	A0	A1 DINO	A2 So400m
LIBERO-10	94.7	94.3	92.7
Goal	96.7	97.3	97.3
Object	98.7	100.0	100.0
Spatial	99.7	99.3	98.7
Four-suite macro	97.42	97.75	97.18

The aggregate deltas are +0.33 and -0.24 points. Thirty of forty A0 tasks already exceed 97%, so this table is a saturation/boundary diagnostic rather than primary support for MINT-VLA. The task-type split observed inside LIBERO-10 (ordered closures positive, unordered collection negative) is exploratory and is tested prospectively in Q4 rather than promoted from these small cells.

A.10 EFFICIENCY AND REPRODUCIBILITY

Parameter trees constructed from the resolved Flax NNX models give the exact counts below. A3’s two 2048×2048 linear layers add 8.39M MACs (approximately 16.78 MFLOPs under the two-FLOP convention) per predictor call, excluding the small attention cost of one additional prefix token.

Arm	Parameters	Added vs. A0	Relative
A0	3,353,433,872	—	—
A1	3,355,008,784	1,574,912	0.047%
A2	3,355,795,216	2,361,344	0.070%
A3	3,361,826,576	8,392,704	0.250%

Under the matched four-A100 training protocol, A0, A2-Abs, and A3 peak at 67,247, 67,247, and 67,265 MiB, respectively; A3 adds 18 MiB (0.027%). Their paired core inference latencies are 57.19, 58.35, and 57.99 ms, and end-to-end WebSocket latencies are 107.20, 108.91, and 107.01 ms. Corresponding throughputs are 9.33, 9.18, and 9.34 requests/s. These measurements show no material A3 deployment penalty relative to A0; they do not establish a speed advantage over other world-model-augmented policies.

The implementation environment pins Python 3.11, JAX 0.5.3 with CUDA 12, Flax 0.10.2, and PyTorch 2.7.1. The audited workspace revision is 8d28087d2f69d063...2d01270e; because the experiment tree contains uncommitted research changes, file hashes above are the authoritative identifiers. Pilot rollout seed groups start at 100k, 200k, 300k, and 400k. The confirmatory launch manifests, reports, and paired scene audit are recorded with the accepted checkpoints and summarized in RESULTS_pi05_confirmatory_3seed_2026-08-02.md. The AI-use disclosure appears in the main paper.